

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

Chapter 11: Convex Minimization Problems

Based on Bauschke & Combettes

July 14, 2026

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Notation You'll Need / Recap I

Ambient space. $\mathcal{H}$ is a real Hilbert space with inner product $\langle \cdot | \cdot \rangle$ and norm $\|\cdot\|$.

Extended-real-valued functions. $f : \mathcal{H} \rightarrow]-\infty, +\infty]$.

- $\text{dom } f = \{x \in \mathcal{H} : f(x) < +\infty\}$ (the *domain*).
- f is *proper* if $\text{dom } f \neq \emptyset$ (it never takes the value $-\infty$, and is finite somewhere).
- $f \in \Gamma_0(\mathcal{H})$ means: f is proper, convex, and lower semicontinuous (lsc) – the standing assumption in most existence results (recap from Chapters 8–9).
- ι_C is the *indicator function* of a set C : 0 on C , $+\infty$ off C . Minimizing $f + \iota_C$ over $\mathcal{H}$ is the same as minimizing f over C .

Level sets. For $\xi \in \mathbb{R}$, $\text{lev}_{\leq \xi} f = \{x \in \mathcal{H} : f(x) \leq \xi\}$ is the *lower level set* at height ξ .

Balls. $B(x; \rho) = \{y \in \mathcal{H} : \|y - x\| \leq \rho\}$.

Quasiconvexity (recap, Ch. 10). f is *quasiconvex* if every lower level set $\text{lev}_{\leq \xi} f$ is convex; *strictly quasiconvex* strengthens this so that no two distinct points on the same level set can have their midpoint at a strictly lower level (this rules out flat regions below the top of a level set).

Strict / strong convexity (recap, Ch. 8–10). f is *strictly convex* if

$f(\alpha x + (1 - \alpha)y) < \alpha f(x) + (1 - \alpha)f(y)$ for all $x \neq y$ in $\text{dom } f$ and $\alpha \in]0, 1[$. f is β -*strongly convex*

Notation You'll Need / Recap II

$(\beta > 0)$ if $f - \frac{\beta}{2} \|\cdot\|^2$ is convex – strictly stronger than strict convexity, and it is what buys us quantitative (Lipschitz-type) control on minimizing sequences, not just uniqueness.

Notation You'll Need / Recap III

What this chapter is about. Given $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ and a subset $C \subseteq \mathcal{H}$, we study the problem

$$\text{minimize } f(x) \text{ subject to } x \in C.$$

Three separate questions organize the chapter:

- ① **Value:** what is $\inf f(C)$, the greatest lower bound on f over C (Section 11.1)?
- ② **Attainment:** is there an actual point $x \in C$ with $f(x) = \inf f(C)$ – a *minimizer* (Section 11.2), and if so is it *unique* (Section 11.3) or does it *exist at all* (Section 11.4)?
- ③ **Approximation:** if no minimizer is at hand yet, how do *minimizing sequences* (sequences whose values approach the infimum) behave – do they converge, and to what (Section 11.5)?

Running numeric example (preview). Throughout, we will track $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto (x - 3)^2 + 1$, minimized first over all of $\mathbb{R}$, then over the restricted convex set $C = [0, 2]$. This one concrete function will illustrate every abstract statement below.

11.1 Intuition: Infimum vs. Minimum

Plain language, first. For $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ and $C \subseteq \mathcal{H}$, write $f(C) = \{f(x) : x \in C\} \subseteq]-\infty, +\infty]$, a set of extended real numbers.

The infimum – a number, not necessarily a point

$\inf f(C)$ is the **greatest lower bound** of the set of values $f(C)$. It always exists in $[-\infty, +\infty]$ (by convention $\inf \emptyset = +\infty$). It only tells you *how low f can get on C* – it says nothing about whether some actual $x \in C$ achieves that value.

The key distinction

Not every infimum is attained. E.g. $f(x) = e^{-x}$ on $\mathbb{R}$ has $\inf f(\mathbb{R}) = 0$, but $f(x) > 0$ for every $x \in \mathbb{R}$: no point achieves the value 0. A **minimizer** is a *point* $x \in C$ with $f(x) = \inf f(C)$ exactly – when a minimizer exists, we write $\min f(C)$ instead of $\inf f(C)$, signaling attainment. Section 11.2 defines minimizers precisely; Sections 11.3–11.4 ask when they are unique / exist at all.

Dually, $\sup f(C)$ is the least upper bound of $f(C)$; everything below has an obvious “upper” mirror image, so we focus on infima (minimization), which is the book’s convention.

11.1 A Continuity Fact About $\inf f(C)$

Why this matters. It is tempting to think that enlarging C to its closure $\overline{C}$ can only *decrease* the infimum (more points to search over). The following shows that, for a convex f , this cannot happen once C has nonempty interior meeting $\text{dom } f$: closing up C does not help.

Proposition 11.1(iv)

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex, let $C \subseteq \mathcal{H}$, and suppose $\text{dom } f \cap \text{int } C \neq \emptyset$. Then

$$\inf f(\overline{C}) = \inf f(C).$$

Proof. $\overline{C} \supseteq C$ gives $\inf f(\overline{C}) \leq \inf f(C)$ trivially (a bigger feasible set can only make the infimum smaller or equal). For the reverse: fix $x_0 \in \overline{C}$, $x_1 \in \text{dom } f \cap \text{int } C$, and set $x_\alpha = (1 - \alpha)x_0 + \alpha x_1$ for $\alpha \in]0, 1[$. Convexity along the segment plus continuity of f relative to its domain near $x_1 \in \text{int } C$ (via Chapter 9's continuity results) give $f(x_0) = \lim_{\alpha \downarrow 0} f(x_\alpha) \geq \inf f(C)$, since each $x_\alpha \in C$ for small α . As $x_0 \in \overline{C}$ was arbitrary, $\inf f(\overline{C}) \geq \inf f(C)$. ■

11.1 Example 11.2: Support Functions Live in Γ_0

Why this example. It shows the infimum/supremum machinery of this section in action: a support function is literally defined as a supremum, and turns out to automatically inherit convexity and lower semicontinuity.

Support function (recap, Ch. 7) and Example 11.2

For $C \subseteq \mathcal{H}$ nonempty, the support function is $\sigma_C(x) = \sup_{y \in C} \langle x \mid y \rangle$. Then:

- σ_C is *sublinear* (positively homogeneous and subadditive) and belongs to $\Gamma_0(\mathcal{H})$;
- $\sigma_C = \sigma_{\overline{\text{conv}} C}$ (support functions cannot distinguish C from its closed convex hull), and $\text{dom } \sigma_C = \text{bar } \overline{\text{conv}} C$ (the *barrier cone*);
- if C is bounded, σ_C is real-valued and continuous on all of $\mathcal{H}$.

Why it works. σ_C is a supremum of the family of continuous linear (hence convex, lsc) functionals $(\langle x \mid \cdot \rangle)_{x \in C}$; a supremum of convex lsc functions is again convex and lsc (Ch. 9), and $\sigma_C(0) = 0$ pins down that it is sublinear rather than just convex.

11.2 Intuition: Local vs. Global for Convex Functions

The punchline of this section, in one sentence. For convex functions, there is no such thing as a “merely local” minimum that fails to be global – convexity kills the usual calculus worry about getting stuck in a shallow local dip.

Why this is special. For a general (nonconvex) function, local minimizers abound and are usually not global. Convexity prevents this: any point that looks locally optimal (best in some small ball around it) must in fact be optimal everywhere. This is what makes convex minimization tractable.

11.2 Definition 11.3: Minimizers, Argmin, Local Minimizers

Definition 11.3

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and let $x \in \mathcal{H}$. Then x is a (*global*) *minimizer* of f if $f(x) = \inf f(\mathcal{H})$, i.e. $f(x) = \min f(\mathcal{H}) \in \mathbb{R}$. The set of minimizers of f is denoted $\text{Argmin } f$. If $\text{Argmin } f$ is a singleton, its unique element is denoted $\text{argmin}_{x \in \mathcal{H}} f(x)$.

Now let $C \subseteq \mathcal{H}$ with $C \cap \text{dom } f \neq \emptyset$. A *minimizer of f over C* is a minimizer of $f + \iota_C$; the set of such minimizers is $\text{Argmin}_C f$. If there exists $\rho \in \mathbb{R}_{++}$ such that x is a minimizer of f over $B(x; \rho)$, then x is a *local minimizer* of f .

- $\text{Argmin } f$: the (possibly empty, possibly non-singleton) set of *global* minimizers.
- $\text{Argmin}_C f$: replace “everywhere” by “on C ” – via the indicator trick $f + \iota_C$, constrained minimization is just unconstrained minimization of a modified function.
- Local minimizer: optimal only within some small ball – a priori a weaker notion than global minimizer.

11.2 Proposition 11.4: Local $\Rightarrow$ Global (Convex Case)

Proposition 11.4

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex. Then every local minimizer of f is a (global) minimizer.

Step-by-step proof.

- 1 Take $x \in \mathcal{H}$ and $\rho \in \mathbb{R}_{++}$ such that $f(x) = \min f(B(x; \rho))$ (i.e. x is a local minimizer, witnessed by the ball of radius ρ).
- 2 Fix any $y \in \text{dom } f \setminus B(x; \rho)$ (a candidate point outside the ball – if no such y exists, x is trivially global). Set $\alpha = 1 - \rho / \|x - y\| \in]0, 1[$ and $z = \alpha x + (1 - \alpha)y$.
- 3 **Key geometric fact:** $z \in B(x; \rho)$. Indeed $\|z - x\| = (1 - \alpha) \|y - x\| = \rho$, so z sits exactly on the boundary of the ball – reachable from x along the segment to y , but not further than ρ .
- 4 By convexity of f along the segment from x to y ,

$$f(x) \leq f(z) = f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y).$$

The first inequality is local optimality at x (since $z \in B(x; \rho)$); the second is convexity.

- 5 Rearranging: $(1 - \alpha)f(x) \leq (1 - \alpha)f(y)$, and since $1 - \alpha > 0$, we get $f(x) \leq f(y)$.
- 6 Since $y \in \text{dom } f \setminus B(x; \rho)$ was arbitrary, and $f(x) \leq f(\cdot)$ already holds trivially inside $B(x; \rho)$ by local optimality, $f(x) \leq f(y)$ for every $y \in \text{dom } f$: x is a global minimizer. ■

11.2 Proposition 11.5: Interior Minimizers Over C Are Global

Proposition 11.5

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex, and let $C \subseteq \mathcal{H}$. Suppose that x is a minimizer of f over C such that $x \in \text{int } C$. Then x is a minimizer of f .

Proof. Since $x \in \text{int } C$, there exists $\rho \in \mathbb{R}_{++}$ with $B(x; \rho) \subseteq C$. Because x minimizes f over C , $f(x) = \inf f(C) \leq \inf f(B(x; \rho))$, and since $B(x; \rho) \subseteq C$ the reverse inequality also holds, so $f(x) = \inf f(B(x; \rho))$: x is a *local* minimizer of f . Proposition 11.4 then upgrades this to a global minimizer. ■

Reading it. This is exactly why, in the running example below, checking whether the constrained minimizer over $C = [0, 2]$ lies in the *interior* of C tells you immediately whether it is also the unconstrained minimizer.

Running Example: Setup

The function. Let $\mathcal{H} = \mathbb{R}$ and

$$f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto (x - 3)^2 + 1.$$

This f is real-valued, continuous, and *strongly convex* with constant $\beta = 2$ (since $f - \frac{1}{2}(2) \|\cdot\|^2$ reduces to an affine function, which is convex).

Minimizing over $\mathcal{H} = \mathbb{R}$ (unconstrained)

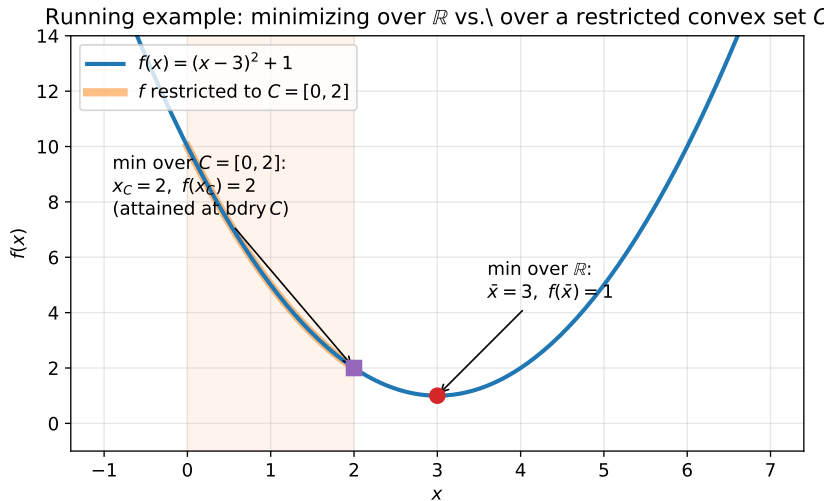
Completing the square is already done: $f(x) \geq 1$ for all $x \in \mathbb{R}$, with equality iff $x = 3$. Hence

$$\operatorname{Argmin} f = \{3\}, \quad \operatorname{argmin}_{x \in \mathbb{R}} f(x) = 3, \quad \min f(\mathbb{R}) = 1.$$

Now restrict. Let $C = [0, 2] \subset \mathbb{R}$: closed, convex, bounded, and $3 \notin C$. What is $\operatorname{Argmin}_C f$?

- On $C = [0, 2]$, f is *strictly decreasing* (since the vertex $x = 3$ lies to the right of all of C), so the minimum over C is attained at the *right endpoint*.
- $x_C = 2$, $f(x_C) = (2 - 3)^2 + 1 = 2$. So $\operatorname{Argmin}_C f = \{2\}$, $\min_{x \in C} f(x) = 2$.
- Since $x_C = 2 \in \operatorname{bdry} C$ (*not* $\operatorname{int} C =]0, 2[$), Proposition 11.5 does *not* apply – consistent with $x_C = 2 \neq 3 = \operatorname{argmin}_{\mathbb{R}} f$: the constrained minimizer is genuinely different from the unconstrained one.

Running Example: Picture



Minimizing $f(x) = (x-3)^2 + 1$ over $\mathbb{R}$ gives $\bar{x} = 3, f(\bar{x}) = 1$ (red dot). Restricting to the closed convex set $C = [0, 2]$ (orange band) moves the minimizer to the boundary point $x_C = 2$ with a strictly larger minimum value $f(x_C) = 2$.

11.3 Intuition: What Forces a Minimizer to Be Unique?

The idea. A convex function can have a whole flat “valley” of minimizers (e.g. $g(x) = \max\{0, |x| - 1\}$ has $\text{Argmin } g = [-1, 1]$). Uniqueness requires ruling out flatness at the bottom: either the function itself must be strictly convex, or the feasible set must be strictly convex so it cannot contain a whole segment of near-optimal points.

Two independent routes to uniqueness (this section)

- 1 Make f *strictly convex* (or $f + \iota_C$ strictly quasiconvex) – Proposition 11.8(i), Corollary 11.9.
- 2 Keep f merely convex, but make the constraint set C *strictly convex* and require the interior of C to miss all unconstrained minimizers – Proposition 11.8(ii).

11.3 Propositions 11.6–11.7: Building Blocks

Proposition 11.6

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and quasiconvex. Then $\text{Argmin } f$ is convex.

Why: $\text{Argmin } f$ is precisely the lowest-height lower level set of f , and quasiconvexity says every lower level set is convex.

Proposition 11.7 (even, convex $\phi : \mathbb{R} \rightarrow]-\infty, +\infty]$)

Suppose ϕ is proper, even ($\phi(-\xi) = \phi(\xi)$), and convex. Then: (i) $0 \in \text{Argmin } \phi$; (ii) ϕ is increasing on $\mathbb{R}_+$; (iii) ϕ is decreasing on $\mathbb{R}_-$. If moreover ϕ vanishes *only* at 0, then (iv) ϕ is strictly increasing on $\mathbb{R}_+$ and (v) strictly decreasing on $\mathbb{R}_-$.

Why: Symmetry plus convexity forces $\phi(0) \leq \frac{1}{2}\phi(\xi) + \frac{1}{2}\phi(-\xi) = \phi(\xi)$ for every ξ , giving (i); a convex combination argument ($\alpha = \xi/\eta$ for $0 < \xi < \eta$) then gives monotonicity, and strictness upgrades to strict monotonicity once 0 is the *only* zero of ϕ . This is exactly the mechanism behind why $\|\cdot\|^2$ (even, strictly convex, vanishing only at 0) has a unique, strictly increasing-away-from-origin radial profile.

11.3 Proposition 11.8: At Most One Minimizer Over C

Proposition 11.8

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be quasiconvex and let C be a convex subset of $\mathcal{H}$ with $C \cap \text{dom } f \neq \emptyset$. Suppose one of the following holds:

- (i) $f + \iota_C$ is *strictly* quasiconvex.
- (ii) f is convex, $(\text{int } C) \cap \text{Argmin } f = \emptyset$, and C is *strictly* convex, i.e.

$$(\forall x \in C)(\forall y \in C) \quad x \neq y \Rightarrow \frac{x + y}{2} \in \text{int } C.$$

Then f has at most one minimizer over C .

Proof sketch. Assume C is not a singleton, set $\mu = \inf f(C)$, and suppose toward a contradiction that $x \neq y$ in $C \cap \text{dom } f$ both achieve $f(x) = f(y) = \mu$. Since x, y lie in the convex set $C \cap \text{lev}_{\leq \mu} f$, so does the midpoint $z = (x + y)/2$, hence $f(z) = \mu$ too.

- (i) Strict quasiconvexity of $f + \iota_C$ forces $\mu = f(z) < \max\{f(x), f(y)\} = \mu$ – impossible.
- (ii) Strict convexity of C gives $z \in \text{int } C$, and Proposition 11.5 upgrades z (a minimizer over C) to a global minimizer of f : $z \in (\text{int } C) \cap \text{Argmin } f$, contradicting the hypothesis that this set is empty.

Either way, no two distinct minimizers can exist.

11.3 Corollary 11.9: The Case $C = \mathcal{H}$

Corollary 11.9

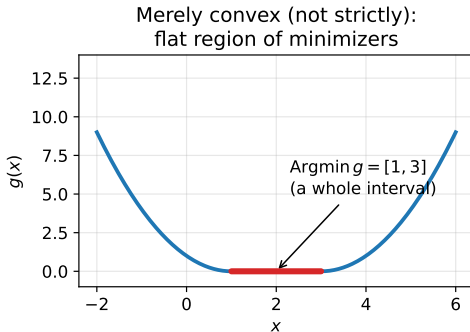
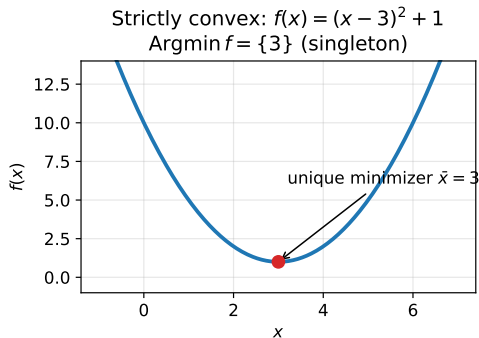
Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and *strictly convex*. Then f has at most one minimizer.

Proof. A strictly convex function is automatically strictly quasiconvex, so Proposition 11.8(i) applies with $C = \mathcal{H}$ (where $\iota_C \equiv 0$, so $f + \iota_C = f$ is strictly quasiconvex). ■

Checking the running example. $f(x) = (x - 3)^2 + 1$ is strictly convex on $\mathbb{R}$ (indeed strongly convex), so Corollary 11.9 guarantees *at most one* minimizer – consistent with $\text{Argmin } f = \{3\}$ found earlier. Uniqueness alone, however, says nothing about *existence*; that is Section 11.4's job.

11.3 Picture: Strict Convexity Is What Buys Uniqueness

Uniqueness of minimizers depends on strict convexity (Corollary 11.9)



Left: $f(x) = (x - 3)^2 + 1$ is strictly convex, so Corollary 11.9 applies – exactly one minimizer, $x = 3$. Right: a merely convex (not strictly convex) function can be perfectly flat — and hence constant at its minimum value — along an entire interval, giving infinitely many minimizers. The distinction is not cosmetic: it is precisely strict convexity that Corollary 11.9 needs.

11.4 Intuition: Why Might a Minimizer Fail to Exist?

Two failure modes to keep in mind.

- **Escape to infinity:** $f(x) = e^{-x}$ on $\mathbb{R}$ has $\inf f = 0$ but no minimizer – the near-optimal points march off to $+\infty$ forever.
- **Boundary not included:** $f(x) = x$ on the open interval $]0, 1[$ has $\inf f = 0$, not attained, because the infimizing point $x = 0$ was excluded from the (non-closed) domain.

The fix, informally. Existence theorems in this section combine three ingredients that jointly rule out both failure modes:

- 1 *Lower semicontinuity* – no sudden downward jumps that would let the infimum “hide” just below a discontinuity;
- 2 *(quasi)convexity* – turns weak convergence into a genuine tool via weak lower semicontinuity (Ch. 10);
- 3 *Boundedness of the relevant lower level set* (via *coercivity*, or a bounded feasible set) – rules out escape to infinity, and combined with closedness and convexity gives *weak compactness* (Ch. 3), on which a continuous (here: lsc) function attains its infimum.

11.4 Theorem 11.10: The Master Existence Theorem

Theorem 11.10

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be lower semicontinuous and quasiconvex, and let C be a closed convex subset of $\mathcal{H}$ such that, for some $\xi \in \mathbb{R}$, $C \cap \text{lev}_{\leq \xi} f$ is nonempty and bounded. Then f has a minimizer over C .

Step-by-step proof.

- 1 Since f is lsc and quasiconvex, Proposition 10.25 (Ch. 10) upgrades this to *weak* lower semicontinuity of f : whenever $x_n \rightharpoonup x$ weakly, $f(x) \leq \liminf f(x_n)$.
- 2 Set $D = C \cap \text{lev}_{\leq \xi} f$. Since C is closed convex and $\text{lev}_{\leq \xi} f$ is closed (by lsc of f) and convex (by quasiconvexity of f), D is closed and convex; by hypothesis D is also nonempty and bounded.
- 3 By Theorem 3.37 (weak compactness of bounded closed convex sets in a Hilbert space), D is *weakly compact*.
- 4 Minimizing f over C is *equivalent* to minimizing f over D : any point of C with f -value $\leq \xi$ lies in D , and any point outside D either leaves C or has f -value $> \xi \geq \inf f(D)$, so cannot be a minimizer over C once $D \neq \emptyset$.
- 5 A weakly lower semicontinuous function attains its infimum on a nonempty weakly compact set (Lemma 2.30 together with the Weierstrass-type Theorem 1.29 applied in $\mathcal{H}^{\text{weak}}$). Hence f attains $\inf f(D) = \inf f(C)$ at some point of $D \subseteq C$: a minimizer of f over C exists. ■

11.4 Coercivity: Definitions 11.11

Intuition. Coercivity is exactly the condition that rules out “escape to infinity”: f must become (and stay) large once you go far enough from the origin, in every direction.

Definition 11.11

Let $f : \mathcal{H} \rightarrow [-\infty, +\infty]$. Then f is *coercive* if

$$\lim_{\|x\| \rightarrow +\infty} f(x) = +\infty,$$

and *supercoercive* if

$$\lim_{\|x\| \rightarrow +\infty} \frac{f(x)}{\|x\|} = +\infty.$$

By convention f is coercive and supercoercive if $\mathcal{H} = \{0\}$.

- Coercive: f eventually exceeds every fixed height ξ as $\|x\| \rightarrow \infty$ – e.g. $f(x) = \|x\|$ is coercive but not supercoercive (grows only linearly).
- Supercoercive: f grows *faster than linearly* in norm – e.g. $f(x) = \|x\|^2$ is supercoercive. Every supercoercive function is coercive, not conversely.

11.4 Propositions 11.12–11.14: Coercivity Toolkit

Proposition 11.12

$f : \mathcal{H} \rightarrow [-\infty, +\infty]$ is coercive $\iff$ all its lower level sets $(\text{lev}_{\leq \xi} f)_{\xi \in \mathbb{R}}$ are bounded.

Why: if some level set is unbounded, you can find a sequence with $\|x_n\| \rightarrow \infty$ but $f(x_n)$ staying bounded – exactly a failure of coercivity, and conversely.

Proposition 11.13 (finite-dimensional refinement)

If $\mathcal{H}$ is finite-dimensional and $f \in \Gamma_0(\mathcal{H})$, then f is coercive $\iff$ *some single* level set $\text{lev}_{\leq \xi} f$ is nonempty and bounded (no need to check *all* of them).

This uses recession cones (Ch. 6): if one bounded level set exists but a higher one is unbounded, the recession cone of the higher level set would be a nonzero direction along which f stays bounded when added to points of the lower level set – contradicting its boundedness. This finite-dimensional shortcut can fail in infinite dimensions (Exercise 11.12).

Proposition 11.14

If $f \in \Gamma_0(\mathcal{H})$ and $g : \mathcal{H} \rightarrow]-\infty, +\infty]$ is supercoercive, then $f + g$ is supercoercive. (Uses that f is minorized by a continuous affine functional, Theorem 9.20.)

11.4 Proposition 11.15 and Corollary 11.16

Proposition 11.15

Let $f \in \Gamma_0(\mathcal{H})$ and let C be a closed convex subset of $\mathcal{H}$ with $C \cap \text{dom } f \neq \emptyset$. If (i) f is coercive, or (ii) C is bounded, then f has a minimizer over C .

Why: pick $x \in C \cap \text{dom } f$; then $D = C \cap \text{lev}_{\leq f(x)} f$ is nonempty, closed, convex, and bounded (by hypothesis (i) or (ii) via Proposition 11.12), so Theorem 11.10 applies directly.

Corollary 11.16

Let $f, g \in \Gamma_0(\mathcal{H})$ with $\text{dom } f \cap \text{dom } g \neq \emptyset$. Suppose (i) f is supercoercive, or (ii) f is coercive and g is bounded below. Then $f + g$ is coercive and has a minimizer over $\mathcal{H}$. If in addition f or g is strictly convex, then $f + g$ has *exactly one* minimizer.

Why: case (i) is Proposition 11.14 (supercoercive $\Rightarrow$ coercive); case (ii) follows since $(f + g)(x) \geq f(x) + \inf g(\mathcal{H}) \rightarrow +\infty$. Uniqueness then follows from Corollary 11.9 since $f + g$ is strictly convex whenever f or g is.

11.4 Corollary 11.17: Strong Convexity Gives Existence *and* Uniqueness

Corollary 11.17

Let $f \in \Gamma_0(\mathcal{H})$ be strongly convex. Then f is supercoercive and it has *exactly one* minimizer over $\mathcal{H}$.

Step-by-step proof.

- ① By definition of β -strong convexity, $q = \frac{1}{2} \|\cdot\|^2$ and there is $\beta \in \mathbb{R}_{++}$ with $f - \beta q$ convex.
- ② Write $f = \beta q + (f - \beta q)$: a sum of the supercoercive function βq (since $q(x)/\|x\| = \|x\|/2 \rightarrow \infty$) and a function $f - \beta q \in \Gamma_0(\mathcal{H})$ (proper, convex by construction, and lsc since f and q both are and q is continuous).
- ③ Proposition 11.14 then gives: f is supercoercive.
- ④ Supercoercive $\Rightarrow$ coercive, so Corollary 11.16(i) (with $g \equiv 0 \in \Gamma_0(\mathcal{H})$, or directly via Proposition 11.15(i) since $f \in \Gamma_0(\mathcal{H})$ is coercive) gives that f has a minimizer over $\mathcal{H}$.
- ⑤ Strong convexity $\Rightarrow$ strict convexity, so Corollary 11.9 gives *at most one* minimizer.
- ⑥ Combining Steps 4 and 5: f has *exactly one* minimizer. ■

Running example check. $f(x) = (x - 3)^2 + 1$ is strongly convex with $\beta = 2$ on $\mathbb{R}$: Corollary 11.17 alone – without any hand computation – already guarantees a unique minimizer exists. We found it explicitly above: $\bar{x} = 3$.

11.4 Proposition 11.18: The Asymptotic Center (Overview)

Why include this. It is a genuinely useful application of the existence *and* uniqueness machinery above, feeding directly into fixed-point theory (Fejér monotonicity, KM iterations from earlier chapters).

Proposition 11.18 (Asymptotic center), key points

Let $(z_n)_{n \in \mathbb{N}}$ be bounded in $\mathcal{H}$, let C be nonempty closed convex, let $T : C \rightarrow C$ be nonexpansive, and set $f : \mathcal{H} \rightarrow \mathbb{R} : x \mapsto \overline{\lim} \|x - z_n\|^2$. Then:

- f is strongly convex with constant 2, hence (by Cor. 11.17 logic) supercoercive;
- $f + \iota_C$ is strongly convex and supercoercive; its unique minimizer z_C (over C) is the *asymptotic center* of $(z_n)_{n \in \mathbb{N}}$ relative to C ;
- if $z_n \rightharpoonup z$ weakly, then $z_C = P_C z$ (the metric projection of the weak limit);
- if $(z_n)_{n \in \mathbb{N}}$ is Fejér monotone w.r.t. C , then $P_C z_n \rightarrow z_C$;
- if $z_{n+1} = Tz_n$ for all n , or if $z_n - Tz_n \rightarrow 0$, then $z_C \in \text{Fix } T$.

Punchline. Existence + uniqueness of a minimizer (guaranteed abstractly by strong convexity, Cor. 11.17) is enough to *define* a canonical point (the asymptotic center) attached to any bounded sequence – a genuinely new object built entirely from this chapter's tools.

11.4 Corollary 11.19: A Boundedness Equivalence

Corollary 11.19

Let C be nonempty closed convex, $T : C \rightarrow C$ nonexpansive, and for $z_0 \in C$ set $z_{n+1} = Tz_n$. The following are equivalent: (i) $\text{Fix } T \neq \emptyset$; (ii) for every $z_0 \in C$, $(z_n)_{n \in \mathbb{N}}$ is bounded; (iii) for some $z_0 \in C$, $(z_n)_{n \in \mathbb{N}}$ is bounded.

Why: (i) $\Rightarrow$ (ii) is standard Fejér-monotonicity theory (Ch. 5); (ii) $\Rightarrow$ (iii) is trivial; and (iii) $\Rightarrow$ (i) is exactly Proposition 11.18(vi): if the orbit is bounded, the asymptotic center construction manufactures an actual fixed point of T . This is a clean illustration of how an *existence-of-minimizer* statement (Cor. 11.17, hidden inside Prop. 11.18) becomes an *existence-of-fixed-point* statement.

11.5 Intuition: Life Without (Yet) Knowing a Minimizer

Recall (Definition 1.8). A *minimizing sequence* of a proper $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ is any sequence $(x_n)_{n \in \mathbb{N}}$ in $\text{dom } f$ with $f(x_n) \rightarrow \inf f(\mathcal{H})$.

Why this matters practically. Numerical algorithms *produce* minimizing sequences (e.g. gradient steps, proximal iterations) long before – if ever – they land on an exact minimizer. Section 11.5 asks: under what conditions does a minimizing sequence actually *converge*, and does it converge to a minimizer?

Spoiler: things can go wrong in several ways

A minimizing sequence can (a) be unbounded, (b) be bounded but have no weak cluster point behave nicely, (c) converge weakly but not strongly, or (d) converge to a point that fails to be a minimizer at all if the corresponding hypotheses are dropped. The examples below (11.23–11.28) are a guided tour of these pathologies, and Propositions 11.20, 11.21, 11.29–11.31 are the positive results that rule them out under suitable assumptions.

11.5 Propositions 11.20–11.21: First Positive Results

Proposition 11.20

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and coercive. Then every minimizing sequence of f is bounded.

Why: immediate from Definition 1.8 (minimizing sequences eventually enter every lower level set at height close to $\inf f$) plus Proposition 11.12 (coercive $\iff$ bounded level sets).

Proposition 11.21

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper, lsc, and quasiconvex, and let $(x_n)_{n \in \mathbb{N}}$ be a minimizing sequence that converges *weakly* to some $x \in \mathcal{H}$. Then $f(x) = \min f(\mathcal{H})$: the weak limit is automatically a minimizer.

Why: Proposition 10.25 (Ch. 10) gives weak sequential lower semicontinuity of f under these hypotheses, so $\inf f(\mathcal{H}) \leq f(x) \leq \underline{\lim} f(x_n) = \inf f(\mathcal{H})$: squeeze.

The catch these two leave open. Coercivity guarantees *boundedness*, not weak *convergence*; and even when a minimizing sequence does converge weakly, Remark 11.22 below shows the target need not be reached *strongly*.

11.5 Remark 11.22: Strong Convergence Is Delicate

Remark 11.22

Let $f \in \Gamma_0(\mathcal{H})$ and $(x_n)_{n \in \mathbb{N}}$ in $\text{dom } f$.

- (i) If $(x_n)_{n \in \mathbb{N}}$ converges *strongly* to a minimizer x of f , and either $\mathcal{H} = \mathbb{R}$ or $x \in \text{int dom } f$, then $(x_n)_{n \in \mathbb{N}}$ *is* a minimizing sequence (continuity of f near x forces $f(x_n) \rightarrow f(x) = \inf f(\mathcal{H})$).
- (ii) Strong convergence to a minimizer does *not* by itself imply the sequence is minimizing, in general (a counterexample uses the construction behind Example 9.43).
- (iii) In *infinite*-dimensional $\mathcal{H}$, even weak convergence of $(x_n)_{n \in \mathbb{N}}$ to a minimizer x with $x \in \text{int dom } f$ need not make $(x_n)_{n \in \mathbb{N}}$ minimizing: take an orthonormal sequence x_n and $f = \|\cdot\|$; then $x_n \rightharpoonup 0$, $f(0) = 0 = \inf f(\mathcal{H})$, yet $f(x_n) \equiv 1$ for every n .

Moral. Finite-dimensionality and interior-of-domain conditions are doing real work in (i); drop either one and strange things can happen, as (ii)–(iii) show concretely.

11.5 A Guided Tour of Pathologies: Examples 11.23–11.26

Setting for all of these: $\mathcal{H} = \mathbb{R}^2$ unless noted.

Example 11.23 – rate of value convergence vs. distance to Argmin

$f(\xi_1, \xi_2) = \xi_2^2/\xi_1$ on $\xi_1 > 0$, 0 at the origin, $+\infty$ otherwise; $\text{Argmin } f = \mathbb{R}_+ \times \{0\}$. Taking $x_n = ((n+1)^{p+2}, n+1)$ for fixed $p \geq 1$ gives $f(x_n) - \min f = O(1/n^p) \rightarrow 0$ (minimizing), yet the distance from x_n to $\text{Argmin } f$ still $\rightarrow +\infty$: values can converge to the infimum arbitrarily fast while the *iterates* drift farther and farther from the minimizer set.

Example 11.24 – similar drift for a simpler function

$f(\xi_1, \xi_2) = \xi_1 + \|(\xi_1, \xi_2)\|$, $\text{Argmin } f = \mathbb{R}_- \times \{0\}$. With $x_n = (-n, 1)$, $f(x_n) - \min f \rightarrow 0$ but the distance to $\text{Argmin } f$ stays $\equiv 1$ (does not even shrink to 0).

Example 11.26 – bounded but not weakly convergent

$f = \iota_{[-x, x]}$ (for fixed $x \neq 0$), $x_n = (-1)^n x$: a bounded minimizing sequence that has *two* distinct cluster points and hence does not converge weakly at all.

11.5 Examples 11.25 and 11.27: Weak but Not Strong Convergence

Example 11.25

$f(\xi_1, \xi_2) = \xi_1 \ln(\xi_1/\xi_2) - \xi_1 + \xi_2$ on $\mathbb{R}_{++}^2$, ξ_2 on $\{0\} \times \mathbb{R}_+$, $+\infty$ otherwise; $f \in \Gamma_0(\mathcal{H})$, $\min f(\mathcal{H}) = 0$, $\text{Argmin } f = \mathbb{R}_+(1, 1)$. Along $\varepsilon_n \downarrow 0$: $x_n = (\varepsilon_n, \varepsilon_n) \rightarrow (0, 0)$ with $f(x_n) \equiv 0$ (minimizing); but $y_n = (\varepsilon_n, e^{-1/\varepsilon_n}) \rightarrow (0, 0)$ has $f(y_n) \rightarrow 1 \neq 0$, and $z_n = (\varepsilon_n, e^{-1/\varepsilon_n^2}) \rightarrow (0, 0)$ has $f(z_n) \rightarrow +\infty$ – *three* sequences converging to the *same* point of $\text{Argmin } f$, with wildly different limiting values of f .

Example 11.27 – infinite dimensions, orthonormal basis

$\mathcal{H}$ infinite-dimensional separable, $(x_k)_{k \in \mathbb{N}}$ an orthonormal basis, $\omega_k \downarrow 0$, and $f(x) = \sum_k \omega_k \langle x | x_k \rangle^2$: real-valued, continuous, strictly convex, unique minimizer 0, but *not coercive* (rescaling along the basis, $y_n = x_n/\sqrt{\omega_n}$, gives $\|y_n\| \rightarrow \infty$ while $f(y_n) \equiv 1$). The minimizing sequence x_n converges weakly but *not strongly* to 0.

11.5 Example 11.28: Alternating Minimization Can Miss the Target

Example 11.28

On $\mathcal{H} = \mathbb{R}^2$, let $f(\xi_1, \xi_2) = \max\{2\xi_1 - \xi_2, 2\xi_2 - \xi_1\} + \iota_{\mathbb{R}_+^2}(\xi_1, \xi_2)$. Then $f \in \Gamma_0(\mathcal{H})$, coercive, unique minimizer $\bar{x} = (0, 0)$, $\min f(\mathcal{H}) = 0$. Starting from $x_0 \in \mathbb{R}_+ \times \mathbb{R}_{++}$, define $x_{n+1} = (\xi_{1,n+1}, \xi_{2,n+1})$ by first minimizing $f(\cdot, \xi_{2,n})$ over $\mathbb{R}$ to get $\xi_{1,n+1}$, then minimizing $f(\xi_{1,n+1}, \cdot)$ over $\mathbb{R}$ to get $\xi_{2,n+1}$ (*alternating*, coordinate-by-coordinate minimization).

The surprising outcome. For every $n \geq 1$, $x_n = (\xi_{2,0}, \xi_{2,0}) \neq \bar{x}$ and $f(x_n) = \xi_{2,0} \neq \inf f(\mathcal{H})$: the iterates get *stuck* immediately (after one full cycle) at a *non-optimal* point and never move again – alternating minimization, applied naively coordinate-by-coordinate to a merely convex (not separately strictly convex in a useful sense) function, need not converge to the true minimizer, and does not even produce a minimizing sequence here.

Takeaway. Coercivity + convexity + lsc guarantee a minimizer *exists* (Theorem 11.10); they say nothing about whether a *particular algorithm* finds it.

11.5 Proposition 11.29: Weak and Strong Convergence Conditions

Proposition 11.29

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper, lsc, and quasiconvex, and let $(x_n)_{n \in \mathbb{N}}$ be a minimizing sequence such that, for some $\xi \in]\inf f(\mathcal{H}), +\infty[$, $C = \text{lev}_{\leq \xi} f$ is bounded. Then:

- (i) $(x_n)_{n \in \mathbb{N}}$ has a weak sequential cluster point, and every such point is a minimizer of f .
- (ii) If $f + \iota_C$ is *strictly* quasiconvex, f possesses a unique minimizer x and $x_n \rightarrow x$.
- (iii) If $f + \iota_C$ is *uniformly* quasiconvex, f possesses a unique minimizer x and $x_n \rightarrow x$ (strongly).

Proof idea for (iii), the strongest conclusion. Let ϕ be the modulus of uniform quasiconvexity of $f + \iota_C$. A direct estimate along the segment from x_n to the (already known, by (ii)) unique minimizer x gives, for $\alpha \in]0, 1[$,

$$\inf f(\mathcal{H}) + \alpha(1 - \alpha)\phi(\|x_n - x\|) \leq f(\alpha x_n + (1 - \alpha)x) \leq \max\{f(x_n), f(x)\} = f(x_n).$$

Since $f(x_n) \rightarrow f(x) = \inf f(\mathcal{H})$, this forces $\phi(\|x_n - x\|) \rightarrow 0$, and since ϕ vanishes only at 0 and is increasing, $\|x_n - x\| \rightarrow 0$. ■

11.5 Corollary 11.30 and Proposition 11.31

Corollary 11.30 (coercive case, $C = \mathcal{H}$)

Let $f \in \Gamma_0(\mathcal{H})$ be coercive and $(x_n)_{n \in \mathbb{N}}$ a minimizing sequence. Then (i) $(x_n)_{n \in \mathbb{N}}$ has a weak cluster point, every one a minimizer; (ii) if f is strictly convex, f has a unique minimizer x and $x_n \rightarrow x$; (iii) if f is uniformly convex on every bounded subset of $\text{dom } f$, f has a unique minimizer x and $x_n \rightarrow x$.

This is Proposition 11.29 specialized to $C = \mathcal{H}$, using coercivity (Prop. 11.12) in place of an assumed bounded level set.

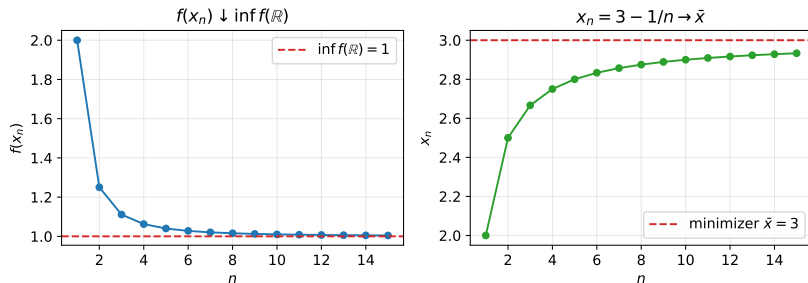
Proposition 11.31 (a genuinely different mechanism: uniform convexity of C , not f)

Let $f \in \Gamma_0(\mathcal{H})$, let C be bounded closed convex with $C \cap \text{dom } f \neq \emptyset$ and $C \cap \text{Argmin } f = \emptyset$, and suppose C itself is *uniformly convex* (a quantitative strengthening of strict convexity of the *set*, not the function). Then any minimizing sequence of $f + \iota_C$ converges *strongly* to the (unique) minimizer of f over C .

Why it is different: here f itself need not be strictly or uniformly convex at all – the strong convergence is bought entirely by geometric uniform convexity of the constraint set C (e.g. any closed ball is uniformly convex, Exercise 11.18).

11.5 Picture: A Well-Behaved Minimizing Sequence

Minimizing sequence for $f(x) = (x - 3)^2 + 1$



For the running example $f(x) = (x - 3)^2 + 1$ (coercive, strictly – indeed strongly – convex on $\mathcal{H} = \mathbb{R}$), take $x_n = 3 - 1/n$. Left: $f(x_n) \downarrow \inf f(\mathbb{R}) = 1$ monotonically – exactly Definition 1.8. Right: $x_n \rightarrow 3 = \operatorname{argmin} f$ strongly, as guaranteed by Corollary 11.30(iii) (strong convexity is uniform convexity on bounded sets). Contrast with Examples 11.23–11.28, where one or more of coercivity, strict/uniform convexity, or lsc is missing, and the nice convergence breaks down.

Python: Minimizers, Minimizing Sequences, and Coercivity

```
import numpy as np

def f(x):
    """Strongly convex running example:  $f(x) = (x-3)^2 + 1$ ."""
    return (x - 3.0) ** 2 + 1.0

# --- Unconstrained minimizer over  $\mathbb{R}$  (Section 11.2/11.4) ---
xs = np.linspace(-50, 50, 2_000_001)
i_star = np.argmin(f(xs))
print("argmin over  $\mathbb{R}$  ~", xs[i_star], " min value ~", f(xs[i_star]))
# argmin over  $\mathbb{R}$  ~ 3.0    min value ~ 1.0    (exact:  $x=3$ ,  $f(x)=1$ )

# --- Constrained minimizer over  $C = [0, 2]$  (Prop. 11.4/11.5) ---
xc = np.linspace(0.0, 2.0, 200_001)
j_star = np.argmin(f(xc))
print("argmin over  $C=[0,2]$  ~", xc[j_star], " min value ~", f(xc[j_star]))
# argmin over  $C=[0,2]$  ~ 2.0    min value ~ 2.0    (attained at bdry  $C$ )

# --- Coercivity check (Definition 11.11 / Proposition 11.12) ---
#  $f$  is coercive iff  $f(x) \rightarrow +\infty$  as  $|x| \rightarrow \infty$ , equivalently
# iff every lower level set  $\{x : f(x) \leq \xi\}$  is bounded.
for xi in [1.5, 5.0, 50.0]:
    level_set = xs[f(xs) <= xi]
    bounded = (level_set.size == 0) or (np.ptp(level_set) < np.inf)
    lo, hi = (level_set.min(), level_set.max()) if level_set.size else (None, None)
    print(f"xi={xi:6.1f}: lev_ $\leq$ xi f = [{lo}, {hi}]    (bounded: {bounded})")
```

Chapter 11 Summary

What we learned

- **11.1 Infima and suprema:** $\inf f(C)$ is always defined, but need not be attained; for convex f with $\text{dom } f \cap \text{int } C \neq \emptyset$, closing up C does not change the infimum (Prop. 11.1(iv)).
- **11.2 Minimizers:** a minimizer is a *point* achieving the infimum (Def. 11.3); for convex f , every *local* minimizer is automatically *global* (Prop. 11.4), and an interior-to- C minimizer over C is a global minimizer of f (Prop. 11.5).
- **11.3 Uniqueness:** strict convexity of f (Cor. 11.9), or strict convexity of C together with $(\text{int } C) \cap \text{Argmin } f = \emptyset$ (Prop. 11.8(ii)), each force at most one minimizer over C .
- **11.4 Existence:** lsc + quasiconvex f with some bounded, nonempty $C \cap \text{lev}_{\leq \xi} f$ guarantees a minimizer over C (Theorem 11.10); coercivity is the key boundedness-of-level-sets ingredient (Prop. 11.12), and *strong* convexity buys both existence and uniqueness at once, via supercoercivity (Cor. 11.17).
- **11.5 Minimizing sequences:** coercivity keeps them bounded (Prop. 11.20); lsc + quasiconvexity makes weak limits automatically minimizers (Prop. 11.21); but Examples 11.23–11.28 show strong/weak convergence and even convergence of the algorithm's iterates to the minimizer set can all fail without extra structure (strict/uniform convexity of f or of C , Props. 11.29–11.31).
- **Running example:** $f(x) = (x - 3)^2 + 1$ has unique minimizer 3 (value 1) over $\mathbb{R}$ and unique